

PH 101
Tutorial 1
Date: 08/08/2023

Q.1-Q.6 are to be discussed in the Tutorial class.

1. (a) If $z = 4x^2 - 8xy^4 + 7y^5 - 3$, find all the first and second order partial derivatives of z .

(b) Let $u = x^2 - 2xy + 2y^3$ with $x = s^2 \ln t$ and $y = 2st^3$. Find $\frac{\partial u}{\partial s}$ and $\frac{\partial u}{\partial t}$.

(c) A cylindrical tank is 1 m high and 0.3 m radius. If height is increased by 5 cm and radius by 1 cm what is the effect on volume? (estimate the increment in volume)

2. The height of a tree increases at a rate of 60 cm per year and the radius increases at 3 cm per year. Find the rate at which the volume of the timber increases when the height is 6 m and the radius is 45 cm. (Assume the tree is a circular cylinder).

3. The height of a hill in **meters** is given by $z = 2xy - 3x^2 - 4y^2 - 18x + 28y + 12$, where x is the distance (in km) east and y is the distance (in km) north of the origin.

(a) Where is the top of the hill and how high is it?

(b) How steep is the slope of the hill at $x = y = 1$? In what direction is the slope steepest, at that point?

4. A particle of mass m is thrown vertically upward with a velocity u from the earth's surface. If the frictional force of air is mkv^2 , where v is the instantaneous velocity and k is the force constant, show that the time taken by the particle to reach the maximum height is given by:

$$T = 1/\sqrt{gk} \tan^{-1}(u\sqrt{k}/\sqrt{g})$$

5. A particle of mass M and charge Q is moving along the x -axis with speed u when it passes between two charged parallel plates. The plates generate a uniform electric field $E_0\hat{j}$ in the region $0 \leq x \leq b$ and no field elsewhere (Refer to Fig. 1).

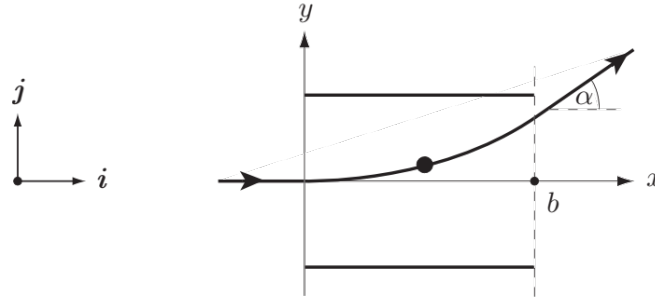


Fig. 1

Find the angle through which the particle is deflected by its passage between the plates.

6. Show that $\vec{\nabla} r^n = n r^{n-1} \hat{r}$, where $r = \sqrt{x^2 + y^2 + z^2}$ and $\hat{r} = \frac{\vec{r}}{r}$,

$$\vec{r} = \hat{x}x + \hat{y}y + \hat{z}z.$$

Practice Problems

7. If the Earth were suddenly stopped in its orbit, how long would it take to collide with the Sun? Assume the Sun as a fixed point mass and also consider that the motion takes place in a straight line through the Sun. [Given: Mass of Sun, $M=2 \times 10^{30} \text{ kg}$; Radius of earth's orbit about the Sun, $R = 1.5 \times 10^{11} \text{ m}$; $G=6.67 \times 10^{-11} \text{ Nm}^2/\text{kg}^2$]

8. A boat of mass M was travelling with velocity u when the boatman stopped rowing. Assuming the retarding force due to water to be $a e^{bv}$ at velocity v (where, a, b are constants), find the time required by the boat to stop. Assume the water to be stationary.

9. Let $u = u(x, y)$ be a function of x and y . Let $x = r \cos \theta$ and $y = r \sin \theta$.

Show that:

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = \frac{\partial^2 u}{\partial r^2} + \frac{1}{r} \frac{\partial u}{\partial r} + \frac{1}{r^2} \frac{\partial^2 u}{\partial \theta^2}$$